



FACULTY OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY
WQD7003 - INTRODUCTION TO DATA ANALYTICS
SAS Assignment
SEMESTER 1 2024/2025

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1.1 Question 1: Describe how to create the pipeline.

Firstly, the breast_data.csv dataset was imported into SAS Enterprise Miner, and classify the variables according to their role and level based on the table below. The ID column is dropped before proceed to the data partition.

Variable	Level	Role
id	Nominal	ID
diagnosis	Binary	Target
radius_mean	Interval	Input
texture_mean	Interval	Input
perimeter_mean	Interval	Input
area_mean	Interval	Input
smoothness_mean	Interval	Input
compactness_mean	Interval	Input
concavity_mean	Interval	Input
concave points_mean	Interval	Input
symmetry_mean	Interval	Input
fractal_dimension_mean	Interval	Input
radius_se	Interval	Input
texture_se	Interval	Input
perimeter_se	Interval	Input
area_se	Interval	Input
smoothness_se	Interval	Input
compactness_se	Interval	Input
concavity_se	Interval	Input
concave points_se	Interval	Input
symmetry_se	Interval	Input
fractal_dimension_se	Interval	Input
radius_worst	Interval	Input
texture_worst	Interval	Input
perimeter_worst	Interval	Input
area_worst	Interval	Input
smoothness_worst	Interval	Input
compactness_worst	Interval	Input
concavity_worst	Interval	Input
concave points_worst	Interval	Input
symmetry_worst	Interval	Input
fractal_dimension_worst	Interval	Input

Data Set Allocations	
Training	50.0
Validation	50.0
Test	0.0

The file then passed through data partitioning. In the data partition step, 50% of the data was used for training, and 50% was used for validation.

Several models with different tuning parameters for decision trees, neural networks, and logistic regression were employed to ensure the model with the lowest misclassification rate and highest predictive accuracy was identified. The specifications of each model are listed in the tables below:

Specification of different models with different tuned:

Model 1

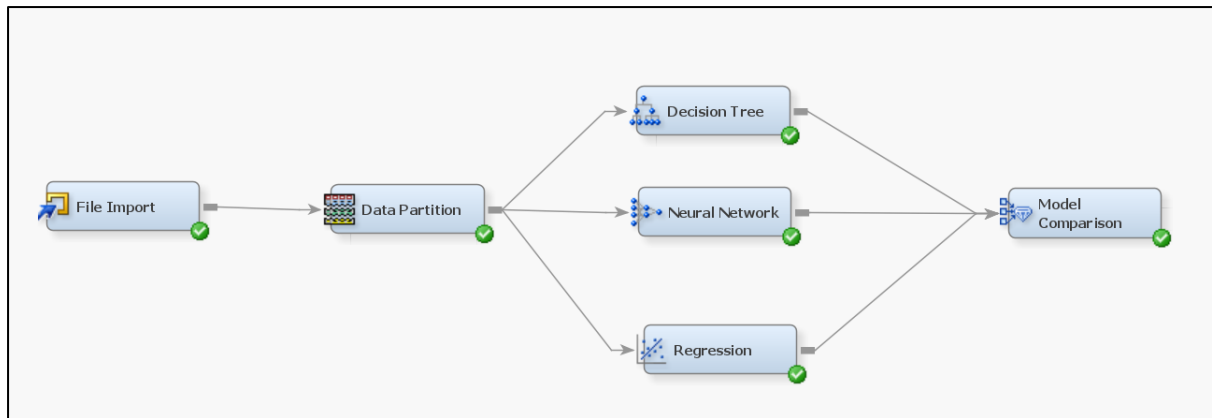
Decision Tree	1	2	3
Maximum Branch	2	3	4
Maximum Depth	6	6	6
Minimum Categorical Size	5	5	5
Leaf Size	5	10	20
Number of Rules	5	5	5
Cross-validation	No	No	Yes

Model 2

Neural Network	1	2	3
Model Selection Criterion	Profit/loss	Misclassification	Average Error
Residuals	Yes	Yes	Yes
Standardization	No	Yes	Yes

Model 3

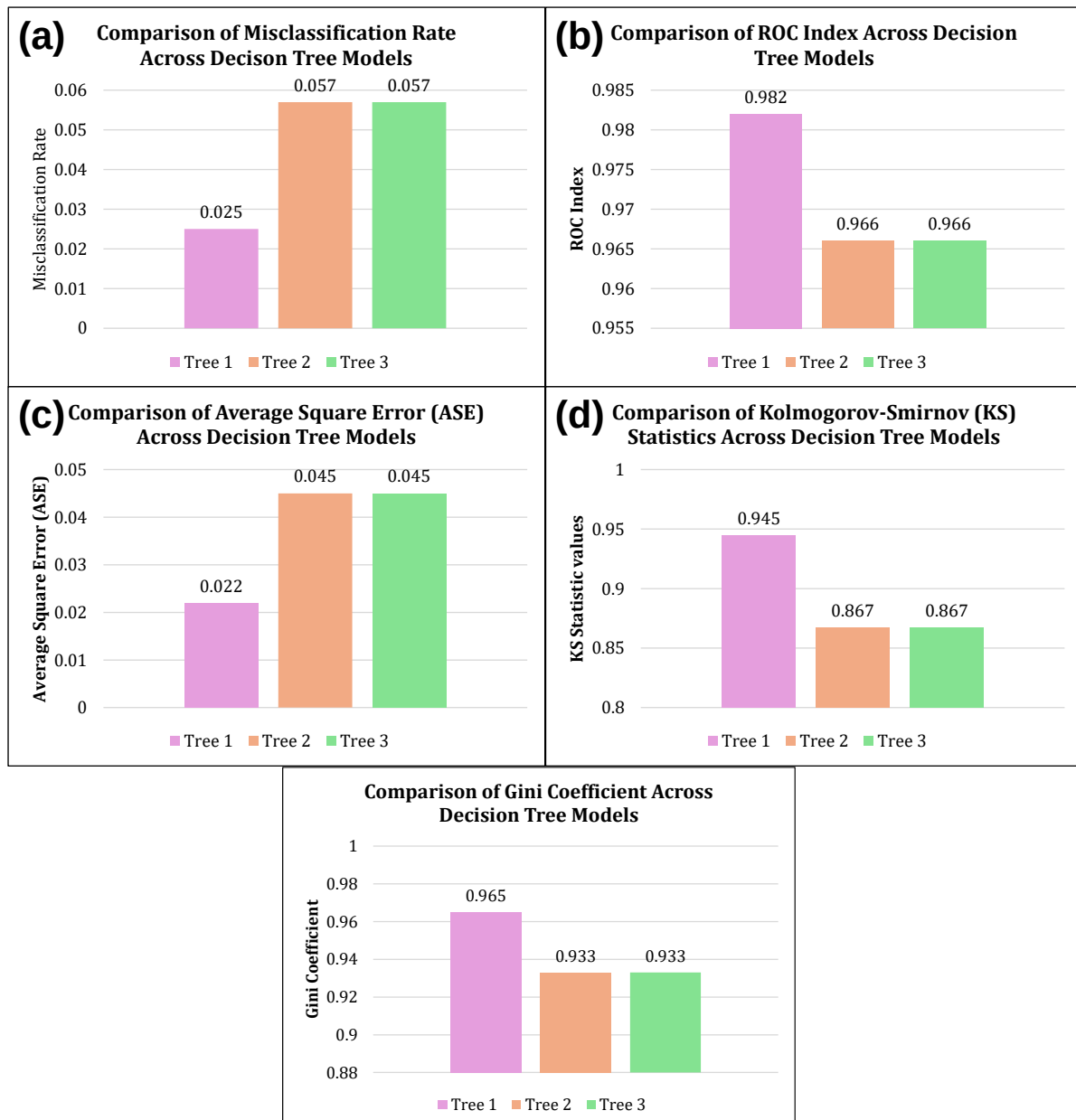
Logistic Regression	1	2	3	4	5	6	7
Main Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Two-Factor Interactions	Yes	Yes	No	No	Yes	Yes	Yes
Selection Model	None	Stepwise	Forward	Backward	Stepwise	Forward	Backward
Selection Criterion	Default	Default	Default	Default	Validation Classification	Validation Classification	Validation Classification



The figure above illustrates the full pipeline of the analysis. This is a simplified version where the process starts with importing a file into SAS then data partitioning. We use three machine learning models (Decision Tree, Neural Network, and Logistic Regression). All models were fine tuned to get the best possible results and maximum accuracy.

1.2 Results

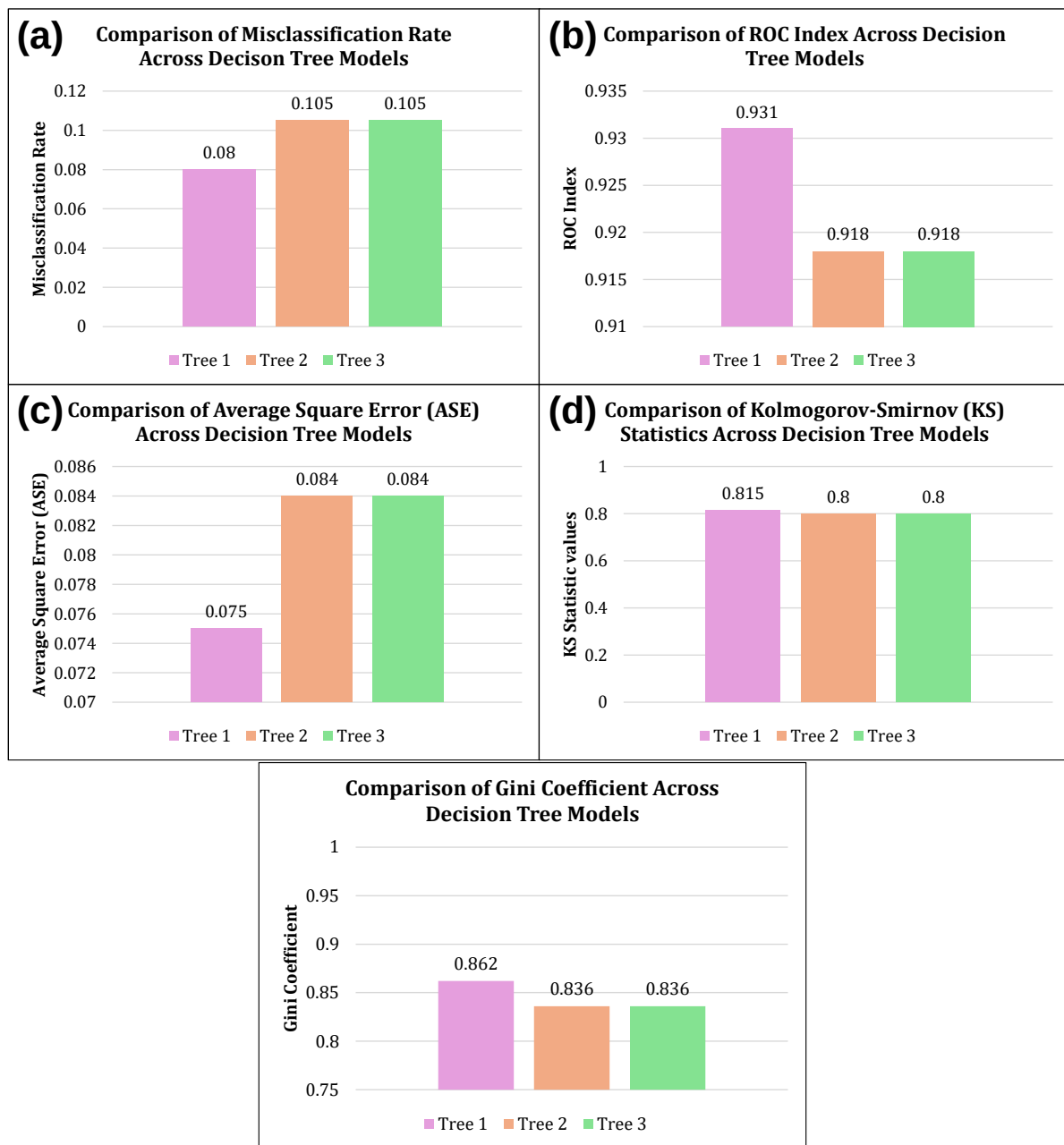
1.2.1 Decision Tree (Train) with different tunes:



For the training data, Tree 1 outperforms the other models with a significantly lower Misclassification Rate (0.025) compared to Tree 2 and Tree 3 (both at 0.057). Its ROC Index (0.982), the highest among the models, indicates excellent classification ability. This superior performance can be attributed to its specification, including the smallest Leaf Size (5) and simplest Maximum Branch (2), which enhance its ability to generalize. Despite lacking cross-validation, Tree 1 achieves a balance of accuracy and interpretability, making it the most effective model configuration. In addition, Tree 1 has the lowest Average Square Error (ASE) (0.022) which indicates its accuracy in

prediction. Additionally, the Kolmogorov-Smirnov (KS) Statistic of 0.945 for Tree 1 confirms that Tree 1 outperforms both Tree 2 (KS Statistic of 0.867) and Tree 3 (Gini Coefficient of 0.933) in distinguishing positive and negative classes.

1.2.2 Decision Tree (Validate) with different tunes

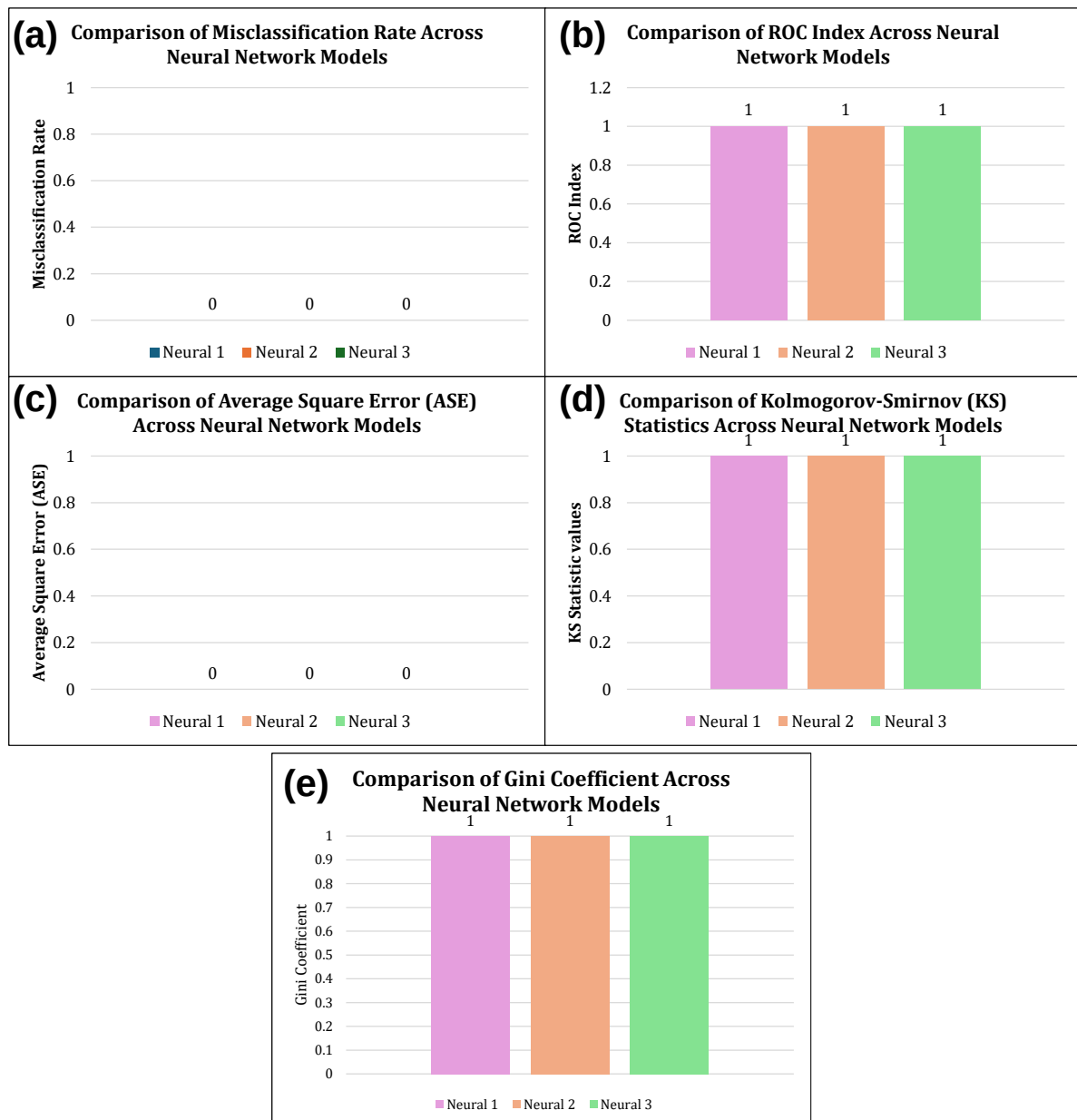


For the validation data, Tree 1 still performs better in terms of the Misclassification Rate (0.08), which is lower than Tree 2 and Tree 3 (both at 0.105). The ROC Index of Tree 1 (0.931) is also higher than the other models' (0.918), indicating better generalization on unseen data. However, its ASE (0.075) is slightly better than Tree 2 and Tree 3's (both at 0.084). The KS Statistic (0.815) and Gini Coefficient (0.862) show that while Tree 1 maintains good performance, the gap is narrower compared to Tree 2 and Tree 3.

Tree 1 concluded as the best-performing model on both the training and validation datasets, excelling in key metrics like Misclassification Rate, ROC Index,

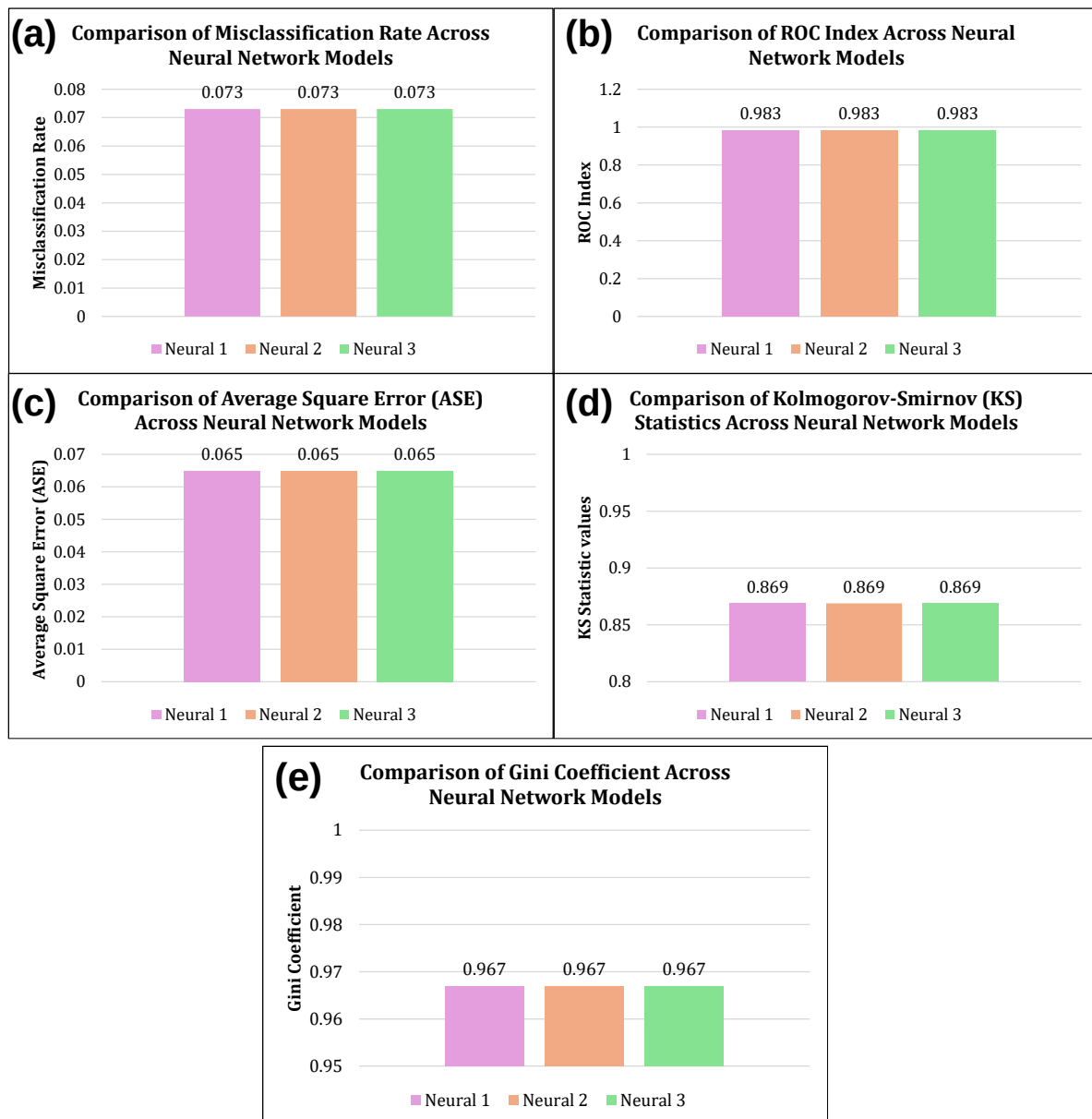
ASE, KS Statistic, and Gini Coefficient. It balances high training performance with good generalization on validation data, making it the most reliable model among the three. Therefore, Decision Tree 1 will be used for comparison with the confusion matrices of other machine learning models.

1.2.3 Neural network (Valid) with different tunes:



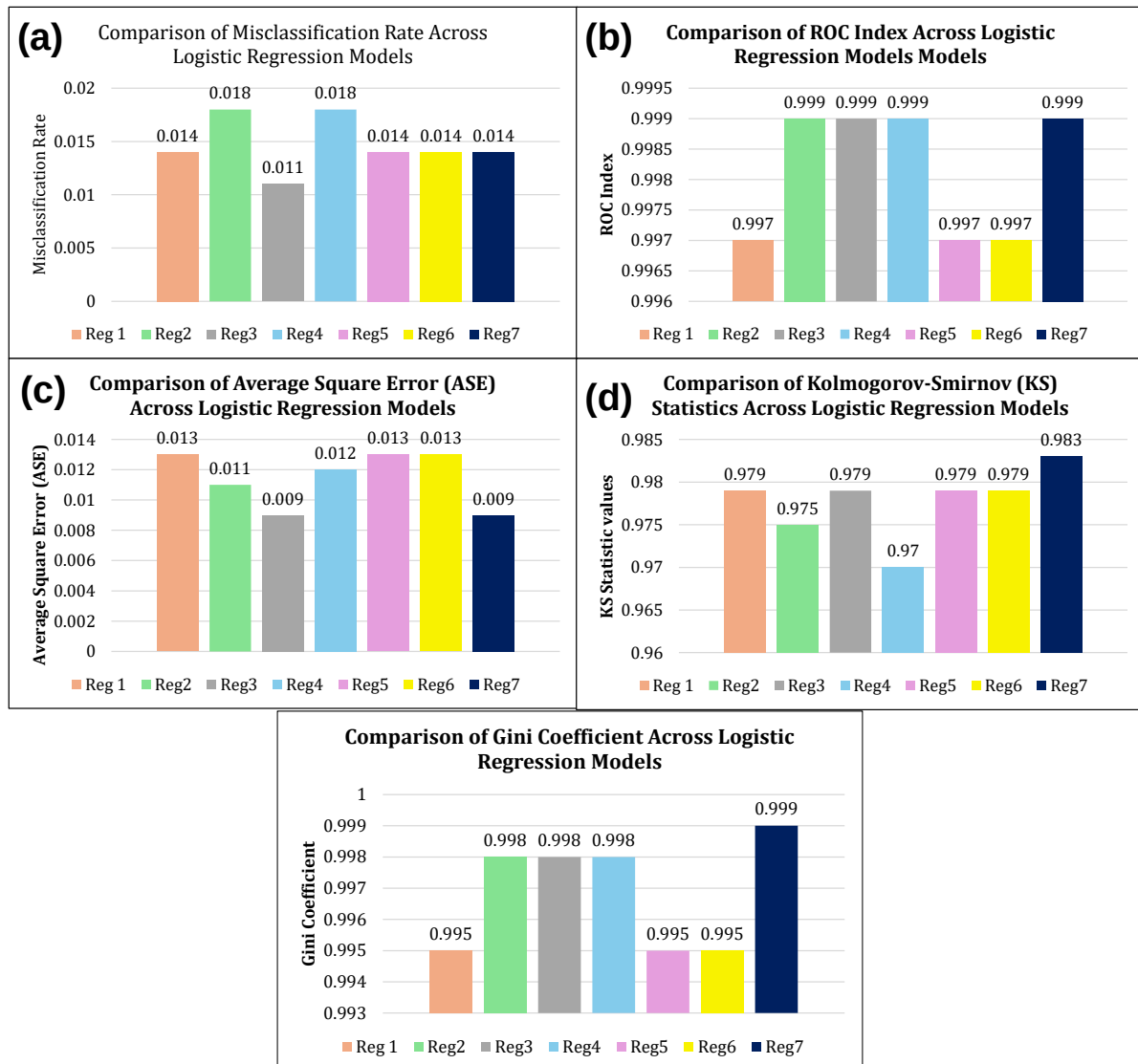
For the training data, Neural 1, Neural 2, and Neural 3 all perform perfectly with Misclassification Rate of 0, ROC Index of 1, ASE of 0, KS Statistic of 1, and Gini Coefficient of 1. This indicates that all three models classify the training data flawlessly without any errors and exhibit the highest possible discrimination capability.

1.2.4 Neural network (Valid) with different tunes:



By comparing Neural Network models (Neural 1, Neural 2, and Neural 3) across different metrics and find that their performance is nearly identical, regardless of tuning variations, indicating a strong consistency in performance of Neural Networks. The Misclassification Rate of all three models is 0.073, which means the same level of accuracy in minimizing errors. Like, their ROC index is at 0.983, which is excellent and consistent classification. All models have an Average Square Error of 0.065, implying that predicted probabilities are very close to the same. Furthermore, the Kolmogorov-Smirnov (KS) Statistic is 0.869 on average for all configurations and the Gini Coefficient is 0.967 for all configurations, confirming the ability of both to differentiate positive and negative classes.

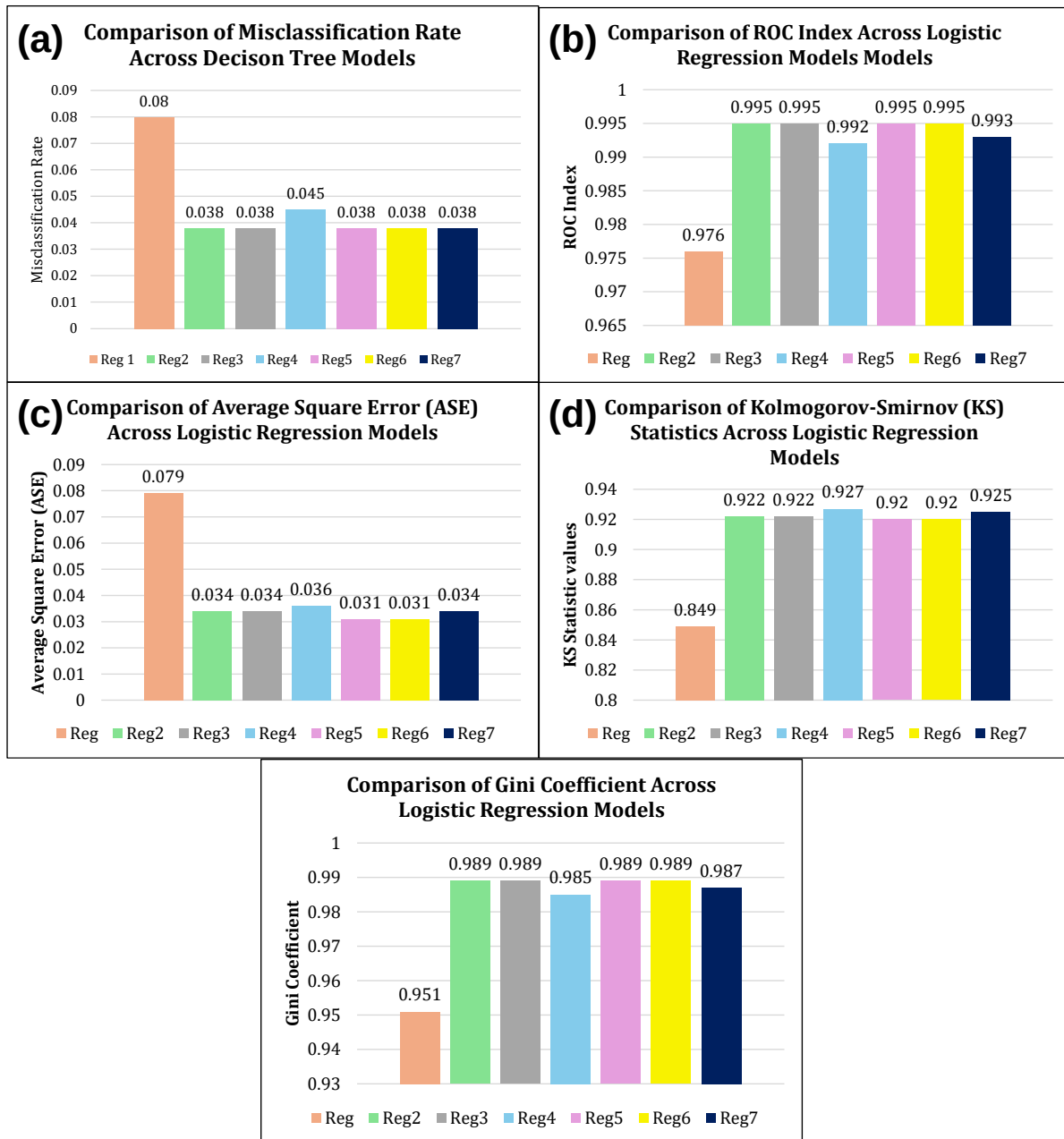
1.2.5 Logistic Regression (Train) with different tunes:



For the training data, Reg 3 stands out as the best model with the lowest Misclassification Rate (0.011) and ASE (0.009). It also achieves a near-perfect ROC Index (0.999), indicating excellent performance in classification tasks. The KS Statistic (0.979) and Gini Coefficient (0.998) for Reg 3 confirm its ability to distinguish between classes effectively.

Other models, such as Reg 2, Reg 4, Reg 5, Reg 6, and Reg 7, also show high performance, with identical ROC Index values (0.999) and Gini Coefficients exceeding 0.997. However, their slightly higher Misclassification Rates (0.014–0.018) and ASEs (0.011–0.013) suggest that Reg 3 is the most accurate on the training dataset.

1.2.6 Logistic Regression (Validate) with different tunes:

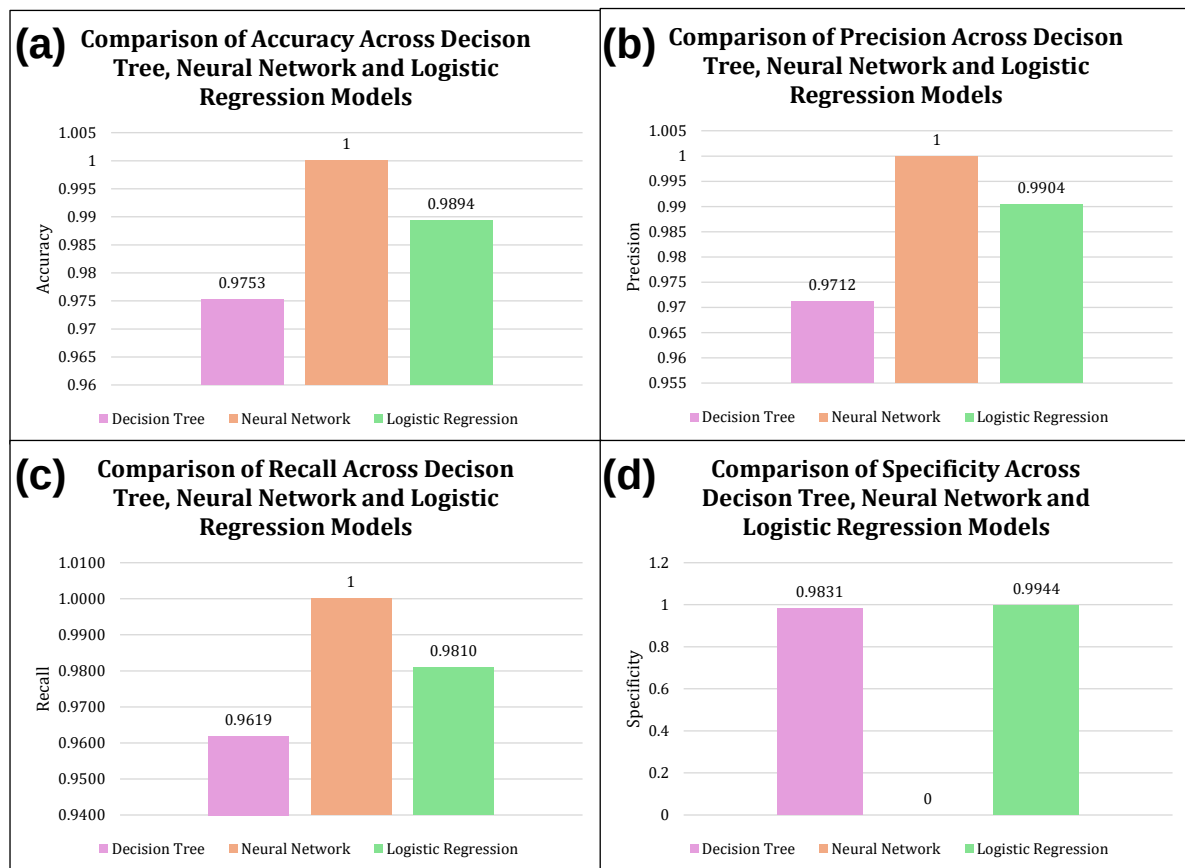


For the validation data, Reg 2, Reg 3, Reg 5, and Reg 6 all achieved a Misclassification Rate of 0.038, which is lower than Reg4 (0.045) and Reg7 (0.038). Among these models, Reg 2, Reg 3, Reg 5, and Reg 6 share identical ASE (0.034), ROC Index (0.995), and Gini Coefficient (0.990), making it difficult to differentiate between them based solely on these metrics.

The KS Statistic provides further insight, with Reg3 and Reg6 achieving slightly higher values (0.922) compared to the other models (0.920). This suggests that Reg3 and Reg6 may have a marginal advantage in class separation on unseen data.

The performance of logistic regression models aligns with their tuning specifications. Reg7, the top performer, includes two-factor interactions, uses backward selection, and applies validation classification, enhancing its accuracy and robustness. In contrast, Reg3, which lacks two-factor interactions, performs well due to its forward selection approach with the default criterion. Models like Reg5 and Reg6, using stepwise and forward selection with validation classification, fall slightly behind, highlighting the importance of optimizing selection methods and interaction terms for better performance.

1.3 Comparison between the three machine learning models (Train)



Based on the comparison of metrics for the training data, the Neural Network demonstrates the strongest performance among the three models: Decision Tree, Neural Network, and Logistic Regression.

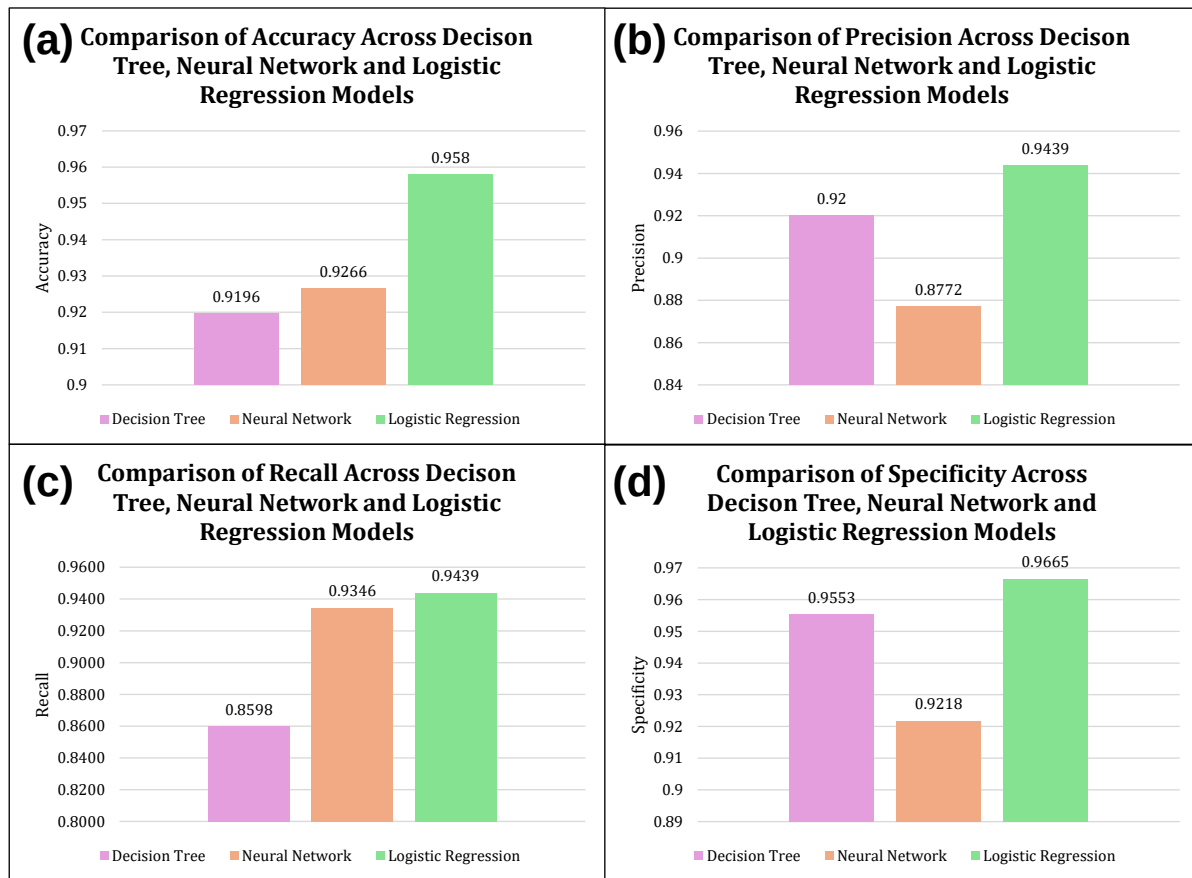
For **Accuracy**, the Neural Network achieves a perfect score of 100%, outperforming Logistic Regression at 98.94% and the Decision Tree at 97.53%.

In terms of **Precision**, the Neural Network again stands out with a perfect score of 100%, significantly surpassing Logistic Regression at 99.04% and the Decision Tree at 97.12%.

Regarding **Recall**, the Neural Network maintains its top position with 100%, highlighting its ability to identify all true positives. Logistic Regression follows at 98.10%, while the Decision Tree lags at 96.19%.

Finally, for **Specificity**, Logistic Regression slightly edges out the Neural Network with a score of 99.44% compared to 98.31% for the Decision Tree. However, the Neural Network's overall dominance across other metrics makes it the most effective model on the training data.

1.4 Comparison between the three machine learning models (Validate)



Based on the comparison across the metrics of Accuracy, Precision, Recall, and Specificity, the Logistic Regression model demonstrates the best overall performance among the three models: Decision Tree, Neural Network, and Logistic Regression.

For **Accuracy**, Logistic Regression achieves the highest score of 95.8%, outperforming the Decision Tree (91.96%) and Neural Network (92.66%), indicating its superior overall classification performance.

Similarly, in terms of **Precision**, Logistic Regression again leads with 94.39%, showing its ability to minimize false positives effectively, while the Decision Tree follows at 92.0%, and the Neural Network lags behind at 87.72%.

When comparing **Recall**, Logistic Regression performs exceptionally well at 94.39%, signifying its ability to correctly identify most true positives, while the Neural Network follows closely with 93.46%. The Decision Tree, however, trails significantly at 85.98%.

Lastly, for **Specificity**, which measures the ability to identify true negatives, Logistic Regression excels with 96.65%, slightly outperforming the Decision Tree at 95.53% and significantly outpacing the Neural Network at 92.18%.

1.5 Question 3: Conclusion on the overall performance based on the statistics

According to the comparison on overall metrics such as accuracy, precision, recall, specificity and others, Logistic Regression seems outperforms the Decision Tree and Neural Network models in most aspects. This shows that it is more accurate to classify positive and negative cases. In addition, its better precision and specificity indicate its capacity to reduce false positives and distinguishing true negatives. While the Neural Network performs stronger on certain metric, like recall, Logistic Regression mostly performs better in overall, albeit balancing better and providing more robust performance across all evaluation criteria. It is the most reliable and complete model of the three with respect to the dataset.

1.6 Appendix

Model 1: Decision Tree

Data Role	Model	Misclassification Rate	ROC Index	ASE	KS Statistic	Gini Coefficient	False Negative	True Negative	False Positive	True Positive	Accuracy	Precision	Recall	Specificity
Train	Tree 1	0.025	0.982	0.022	0.945	0.965	4	175	3	101	0.9753	0.9712	0.9619	0.9831
	Tree 2	0.057	0.966	0.045	0.867	0.933	11	173	5	94	0.9435	0.9495	0.8952	0.9719
	Tree 3	0.057	0.966	0.045	0.867	0.933	11	173	5	94	0.9435	0.9495	0.8952	0.9719
Validate	Tree 1	0.08	0.931	0.075	0.815	0.862	15	171	8	92	0.9196	0.92	0.8598	0.9553
	Tree 2	0.105	0.918	0.084	0.8	0.836	20	169	10	87	0.8951	0.8969	0.8131	0.9441
	Tree 3	0.105	0.918	0.084	0.8	0.836	20	169	10	87	0.8951	0.8969	0.8131	0.9441

Model 2: Neural Network

Data Role	Model	Misclassification Rate	ROC Index	ASE	KS Statistic	Gini Coefficient	False Negative	True Negative	False Positive	True Positive	Accuracy	Precision	Recall	Specificity
Train	Neural 1	0	1	0	1	1	.	178	.	105	1	1	1	N/A
	Neural 2	0	1	0	1	1	.	178	.	105	1	1	1	N/A
	Neural 3	0	1	0	1	1	.	178	.	105	1	1	1	N/A
Validate	Neural 1	0.073	0.983	0.065	0.869	0.967	7	165	14	100	0.9266	0.8772	0.9346	0.9218
	Neural 2	0.073	0.983	0.065	0.869	0.967	7	165	14	100	0.9266	0.8772	0.9346	0.9218
	Neural 3	0.073	0.983	0.065	0.869	0.967	7	165	14	100	0.9266	0.8772	0.9346	0.9218

Model 3: Logistic Regression

Data Role	Model	Misclassification Rate	ROC Index	ASE	KS Statistic	Gini Coefficient	False Negative	True Negative	False Positive	True Positive	Accuracy	Precision	Recall	Specificity
Train	Reg 1	0.014	0.997	0.013	0.979	0.995	.	178	.	105	N/A	N/A	N/A	N/A
	Reg2	0.018	0.999	0.011	0.975	0.998	2	175	3	103	0.9823	0.9717	0.9810	0.9831
	Reg3	0.011	0.999	0.009	0.979	0.998	2	177	1	103	0.9894	0.9904	0.9810	0.9944
	Reg4	0.018	0.999	0.012	0.97	0.998	3	176	2	102	0.9823	0.9808	0.9714	0.9888
	Reg5	0.014	0.997	0.013	0.979	0.995	2	176	2	103	0.9859	0.981	0.9810	0.9888
	Reg6	0.014	0.997	0.013	0.979	0.995	2	176	2	103	0.9859	0.981	0.9810	0.9888
	Reg7	0.014	0.999	0.009	0.983	0.999	2	176	2	103	0.9859	0.981	0.9810	0.9888
Validate	Reg 1	0.08	0.976	0.079	0.849	0.951	10	166	13	97	0.9196	0.8818	0.9065	0.9274
	Reg2	0.038	0.995	0.034	0.922	0.989	6	174	5	101	0.9615	0.9528	0.9439	0.9721
	Reg3	0.038	0.995	0.034	0.922	0.989	6	173	6	101	0.958	0.9439	0.9439	0.9665
	Reg4	0.045	0.992	0.036	0.927	0.985	6	172	7	101	0.9545	0.9352	0.9439	0.9609
	Reg5	0.038	0.995	0.031	0.92	0.989	5	173	6	102	0.9615	0.9444	0.9533	0.9665
	Reg6	0.038	0.995	0.031	0.92	0.989	5	173	6	102	0.9615	0.9444	0.9533	0.9665
	Reg7	0.038	0.993	0.034	0.925	0.987	6	174	5	101	0.9615	0.9528	0.9439	0.9721